MONTCLAIR
STATE UNIVERSITY

Human Resources Data Visualization Analysis

Data Visualization

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Abstract

This project analyzes a workforce dataset consisting of approximately 30,000 employee records to understand demographic representation and distribution patterns across departments, leadership levels, age groups, and geographic regions. Rather than beginning with assumptions regarding where disparities might exist, we approached the dataset with the goal of allowing patterns to emerge through structured visual analysis in Tableau, addressing four research questions which guided our work: (1) where gender gaps exist across departments, (2) whether gender representation differs in managerial or executive roles, (3) whether age distributions differ across the departments or roles and (4) where employees are geographically located.

Across the dataset, gender composition is reasonably balanced (47.2% female and 52.8% male), with only minor variations by department and leadership categories. Employee ages range from 22 to 60 with a median of 41. Geographically, employees are distributed across ten countries, with the largest share located in Germany (10.3% of employees). While the findings suggest demographic consistency across groups, the visualizations developed in this project demonstrate how descriptive analytics can provide organizations with a practical baseline for monitoring representation and supporting informed workplace planning decisions.

Introduction and Motivation

Workforce composition analysis is more than a matter of compliance reporting or simple demographic summaries. Such an analysis directly forms recruiting strategy, leadership development, succession planning and long-term sustainability. This can drive company success as it has been reported that companies with 30% of women executives are found to outperform companies whose proportion is 10%-30%[1]. In recognizing that representation is a practical concern for these workforce strategies, a descriptive demographic analysis provides the structured framework for evaluating representation and identifying areas of imbalance for opportunities of improvement within specific departments, leadership levels or geographic locations. Further, such analyses can provide employers with the ability to monitor progress overtime, and prioritize interventions such as targeted recruiting, leadership development, or retention efforts.

As a team, we approached this project with a data-driven mindset. Our goal was to objectively examine distribution patterns and assess whether meaningful differences exist across organizational groups. Throughout the process, we aimed to produce clear and understandable visuals while still allowing the viewer to explore the relationships independently.

This report focuses on descriptive analysis using Tableau and is organized around four well-defined research questions. The focus of our work is not on establishing cause-and-effect relationships, but on creating dashboard-ready visualizations that clearly highlight patterns of representation. In practice, this type of analysis can serve as a baseline diagnostic tool that organizations can use to monitor demographic trends over time, identify potential areas of concern, and support strategic workforce planning decisions.

Research Questions

Q1. Where do gender gaps exist across departments?

Q2. Are genders represented differently in managerial or executive roles?

Q3. Are there differing patterns in age distribution across roles or departments?

Q4. Where are the majority of employees geographically located?

SMART Goal

Within this project, the goal is to create clear, interactive set of Tableau visuals that enables a viewer to identify any major representation gaps by department and leadership level, compare age distributions across organizational groups and pinpoint the main geographic employee concentrations. Success is defined as producing a dashboard-ready set of charts with readable labels, consistent formatting, and filters that support exploration. The Tableau dashboard has been published for view and interaction here: https://public.tableau.com/app/profile/rachel.tuly/viz/Group1WorkforceDemographicsDashboard/WorkforceDemographicsDashboard

Data Set

The analysis uses the provided employee file employee_records.xlsx, which contains 30,000 employee-level records. Each row represents an individual employee with demographic, job and compensation attributes. No personal identifiers beyond an employee name and employee ID are used in the analysis.

Variables and Data Dictionary

Variable	**Data Type**	**Description**
Employee_ID	Numerical/Categorical	Unique employee identifier
Employee_Name	Categorical	Employee name (Used for display, not analysis)
Age	Numerical	Employee age, in whole years
Gender	Categorical	Gender Category (Male/Female)
Country	Categorical	Country Location
Department	Categorical	Department assignment (Engineering, Finance, HR, Marketing, Sales, Support)
Position	Categorical	Job position (Analyst, Assistant, Consultant, Developer, Executive, Manager)
Salary	Numerical	Annual salary amount (USD)
Joining_Date	Date	Employee hire/joining date

Data Preparation and Quality Checks

Dataset was reviewed for common data quality issues and prepared for Tableau using key steps:

- Verified that the Employee_ID is unique (duplicate IDs found: 0)
- Checked for missing values across all fields (no missing values detected)
- Confirmed appropriate data types (Age and Salary as numeric measures, Joining_Date as date, and Country as geographic role)
- Created calculated fields used in the visuals (Headcount, Leadership Role Level and Age Group)
- Assessed gender based on common applications of names in dataset.
- Applied percent-of-total table calculations where composition comparisons were needed (Q1 to Q3)

Field Name	Missing Values
Employee_ID	0
Employee_Name	0
Age	0
Gender	0
Country	0
Department	0
Position	0
Salary	0
Joining_Date	0

Basic Profile of the Data

Metric	**Value**
Total employees (rows)	30,000
Overall gender split	47.2% female, 52.8% male
Age (min, median, max)	22, 41, 60
Age (mean)	41.0
Number of departments	6
Number of countries	10

Analysis

This project uses descriptive analytics and visualization methods in Tableau. The primary measure used across charts is Headcount, calculated as the distinct count of Employee_ID. Additional calculated fields support grouping and comparison across leadership level and age bands. Charts are formatted as percent-of-total when the goal is to compare composition rather than volume.

Methodology

This analysis combines descriptive statistics, proportional comparisons, and regression-based modeling to evaluate demographic representation across departments and leadership levels. Categorical variables were analyzed using baseline reference groups to determine the modeling software, with coefficients interpreted as deviations from those reference categories while holding other variables constant. Visual outputs were developed in Tableau to translate statistical results into an accessible, decision-ready dashboard.

Key Tableau Calculations Used

Item	Definition
Headcount	COUNTD([Employee ID])
Role Level	IF Position = 'Executive' THEN 'Executive' ELSEIF Position = 'Manager' THEN 'Manager' ELSE 'Non-Leadership' END
Age Group	Under 30, 30-39, 40-49, 50-59, 60+ (created via IF/ELSEIF on Age)
Percent of Total	Quick Table Calculation used to show composition within each Department or Role Level
Top Countries	Country filter limited to Top 10 by Headcount for map readability

Results

Q1. Gender Mix by Department

To evaluate gender gaps in staffing across departments, we used a 100% stacked bar chart visualization showing the percentage of total headcount by gender within each department. This format makes it easy to compare composition across departments, even when department sizes differ. We find that while there are some slight variations between departments, gender is spread roughly equal between each of the groups, though the population of male employees is consistently larger in all departments.

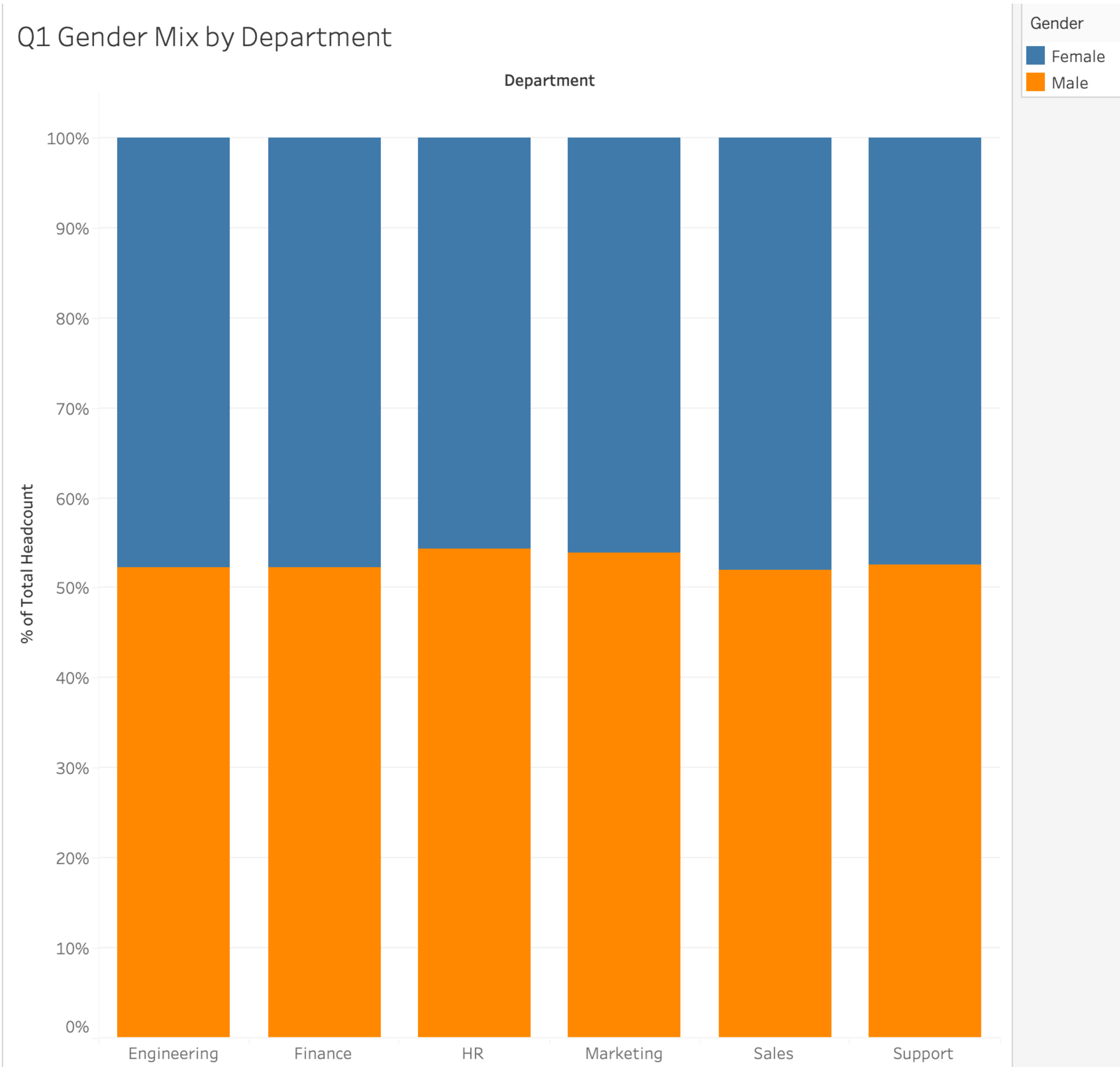

Figure 1. Q1 Gender Mix by Department (percentage of headcount)

Department	Female %	Male %
Engineering	47.8%	52.2%
Finance	47.8%	52.2%
HR	45.7%	54.3%
Marketing	46.1%	53.9%
Sales	48.1%	51.9%
Support	47.5%	52.5%

Q2. Gender Representation by Leadership Level

To assess whether gender representation differs in managerial or executive roles, we grouped position into three Role Levels: Executive, Manager and Non-Leadership. We then used a 100% stacked bar chart to compare gender composition within each role level. As we found with the gender variation across departments, each of the leadership levels demonstrates a slightly higher number of male employees over female employees. While this could be identified as an imbalance and an opportunity for improvement, the proportions are consistent with the company overall, and the proportion is at the lowest point among executive leadership and would therefore not be a cause for concern.

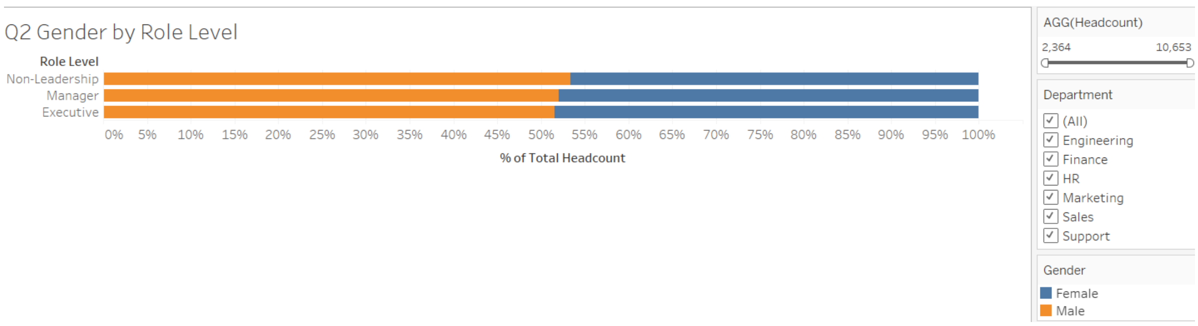

Figure 2. Q2 Gender by Role Level (percent of headcount)

Role Level	Female %	Male %
Executive	48.5%	51.5%
Manager	48.0%	52.0%
Non-Leadership	46.6%	53.4%

Q3. Age Distribution Across Departments and Roles

Age patterns were analyzed using two complementary views. First, an age group composition chart shows how employees in each department are distributed across age bands. Second, a box plot summarizes the age distribution per department using the median and interquartile range. We find that overall, the smallest age group is 60+, with a relatively even distribution between the younger groups. Overall, the employer does not appear at risk for losing a large population of employees in the coming years due to retirement.

Figure 3. Q3 Age Group Composition by Department (percent of headcount).

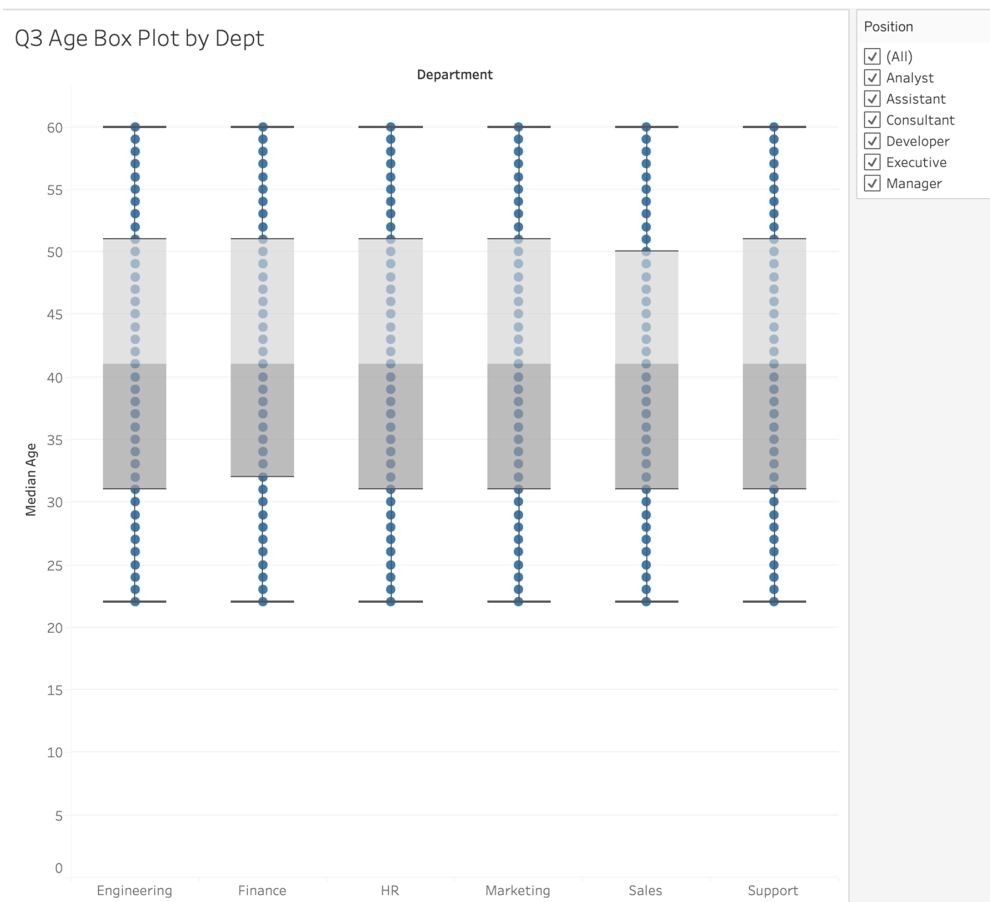

Figure 4. Q3 Age Box Plot by Department (distribution summary).

Department	N	Mean Age	Median Age	Min	Max
Engineering	5070	41.0	41	22	60
Finance	4938	41.2	41	22	60
HR	4976	40.8	41	22	60
Marketing	5007	41.0	41	22	60
Sales	5019	40.9	41	22	60
Support	4990	40.9	41	22	60

Q4. Geographic Distribution of employees

To identify where employees are geographically located, we created a map visualization using Country as the geographic dimension, and headcount as both the size and color. A Top 10 filter by headcount was applied to focus the view and reduce clutter. The country with the highest headcount in the dataset is Germany, with 3,090 employees (10.3%).

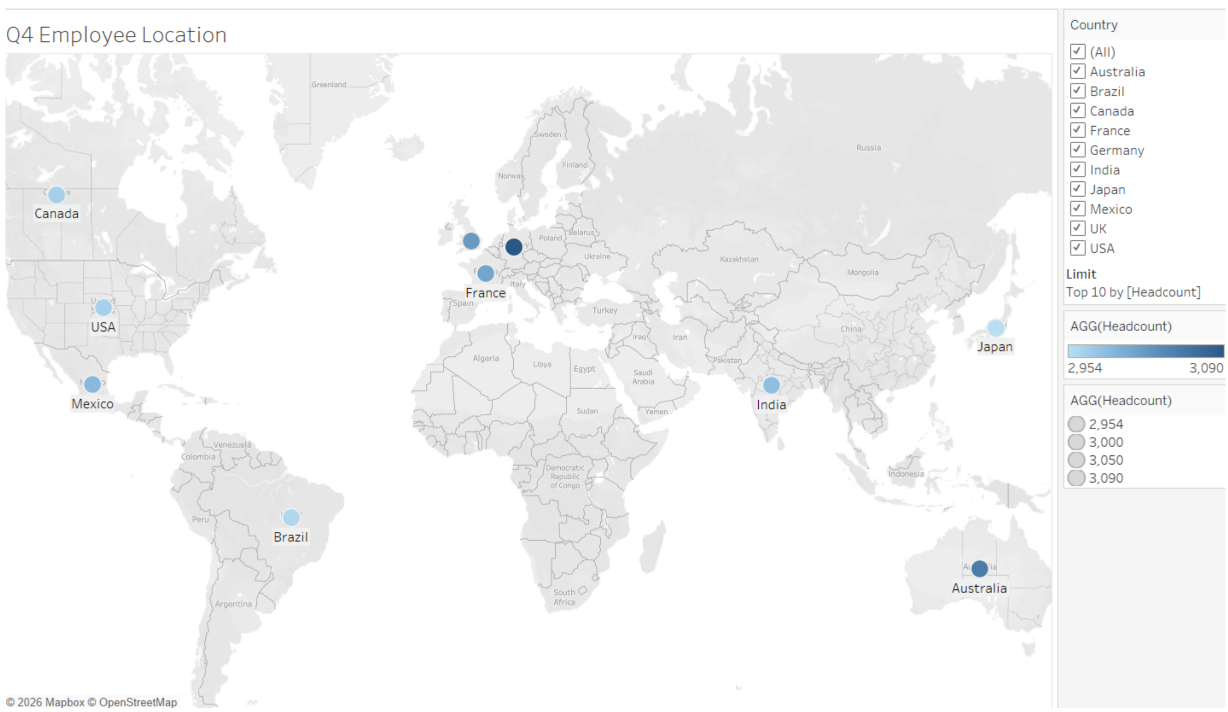

Figure 5. Q4 Employee Location Map (Top Countries by Headcount).

Country	Headcount	Share of Total
Germany	3090	10.3%
Australia	3052	10.2%
UK	3021	10.1%
France	3010	10.0%
Mexico	2991	10.0%
India	2985	10.0%
Canada	2969	9.9%
USA	2967	9.9%
Brazil	2961	9.9%
Japan	2954	9.8%

Additional Exploration: Salary Patterns (Descriptive and Regression

As an additional exploration aligned with course guidance, Salary was treated as a dependent variable, with Age, Gender, Department and Position as predictors. This section summarizes descriptive salary patterns and a simple linear regression model.

Salary Summary by Department

Department	N	Mean	Median	Min	Max
Engineering	5070	90,159.59	90,502.16	30,004.78	149,989.60
Finance	4938	89,526.91	88,854.42	30,019.60	149,841.52
HR	4976	89,930.38	89,622.39	30,021.75	149,995.00
Marketing	5007	89,834.05	89,256.08	30,024.84	149,953.43
Sales	5019	88,854.23	87,830.75	30,017.91	149,996.39
Support	4990	90,649.14	91,096.82	30,048.30	149,971.79

Salary Summary by Position

Position	N	Mean	Median	Min	Max
Analyst	5008	89,909.85	89,196.62	30,021.75	149,989.60
Assistant	4934	90,436.26	91,050.82	30,019.60	149,964.65
Consultant	5022	89,018.68	87,522.87	30,017.91	149,989.17
Developer	4988	89,528.07	89,271.06	30,004.78	149,996.39
Executive	5124	90,604.78	90,608.58	30,053.71	149,979.37
Manager	4924	89,444.87	88,848.53	30,037.83	149,994.40

Regression Overview (OLS)

Model: Salary = f(Age, Gender, Department, Position). $R^2 = 0.0005$, Adjusted $R^2 = 0.0001$. These values are very small, which suggests the available predictors explain very little of the variation in salary in this dataset. Results should be interpreted as exploratory rather than causal.

Term (relative to baseline categories)	Coefficient
C(Department)[T.Sales]	-1,305.70
C(Position)[T.Consultant]	-896.94
C(Position)[T.Executive]	699.59
C(Department)[T.Finance]	-619.42
C(Position)[T.Assistant]	523.67
C(Department)[T.Support]	496.47
C(Position)[T.Manager]	-470.66
C(Position)[T.Developer]	-380.39

Baseline categories are determine by the modeling software and typically reflect the first category (alphabetical) within each field. Coefficients represent differences relative to those baseline categories while holding other variables constant.

Discussion and Practical Implications

Overall, the dataset shows relatively uniform demographic composition across departments and role levels. In practice, this type of analysis can be used as a baseline for monitoring representation over time and for identifying where deeper analysis is needed. If a real organization dataset shows larger differences, leaders could use the results to prioritize recruiting strategies, promotion review, leadership development programs and retention interventions targeted to specific departments or levels. The significance of such efforts in recruiting can be monetarily significant; when considering profits, it has been found that "companies in the top quartile for gender diversity on executive teams were 25 percent more likely to have above-average profitability than companies in the fourth quartile."[1]

Dashboard and Story Telling Plan (Tableau)

A simple story flow for a Tableau dashboard is: (1) start with gender composition by department to identify gaps, (2) move to leadership level to evaluate representation in management and executive roles, (3) examine age distribution to understand workforce mix and potential retirement risk, and (4) end with geography to identify where staffing is concentrated. Filters for Department, Position and country can be applied globally so viewers can explore segments and compare patterns.

Limitations and Conclusion

This project is primarily descriptive and relies on the variables available in the provided dataset. Key limitations include: (1) the analysis does not include tenure outcomes, performance, attrition or hiring and promotion events, which are critical for diagnosing root causes; (2) gender is represented as a binary category in the dataset, which limits inclusivity and detail; (3) results reflect a single snapshot of data, so trends over time cannot be assess to specific policies or practices; (4) an important aspect of diversity is race and ethnicity, which was not included in our data set. Additionally, this analysis does not examine intersectional patterns gender by geography which may reveal disparities not visible in aggregate comparisons. Future work could add longitudinal data, richer demographic categories and modeling approaches to explore predictors of pay, promotion or retention.

In conclusion, the Tableau visuals developed in this project provide a clear, dashboard-ready view of workforce demographics and representation. The charts answer the four research questions by showing gender composition by department and role level, age distribution comparisons and geographic employee concentrations. These outputs can support communication with stakeholders and serve as a starting point for deeper workforce analytics.

References

1. Dixon-Fyle, S., Dolan, K., Hunt, D.V., Prince, S. (2020). *Diversity Wins: How Inclusion Matters*. McKinsey. https://www.mckinsey.com/featured-insights/diversity-and-inclusion/diversity-wins-how-inclusion-matters.
2. Smayan, J. (2024). *Employee Records Dataset* (Version 2) [Data set]. **Kaggle**. https://www.kaggle.com/datasets/smayanj/employee-records-dataset

Appendix: How to Recreate the Visuals in Tableau (Short Steps)

Visual	**Build Steps**
Q1 Gender Mix by Department	Department on Columns, Headcount on Rows, Gender on Color, Percent of Total table calc.
Q2 Gender by Role Level	Role Level on Rows, Headcount on Columns, Gender on Color, Percent of Total table calc, Department filter optional.
Q3 Age Distribution	Age Group on Color and Percent of Total by Department, plus a box plot using Age with Employee ID on Detail.
Q4 Employee Location	Country on map (generated lat/long), Headcount on Color and Size, Top N filter by Headcount.